

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Grasshopper Sparrow"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

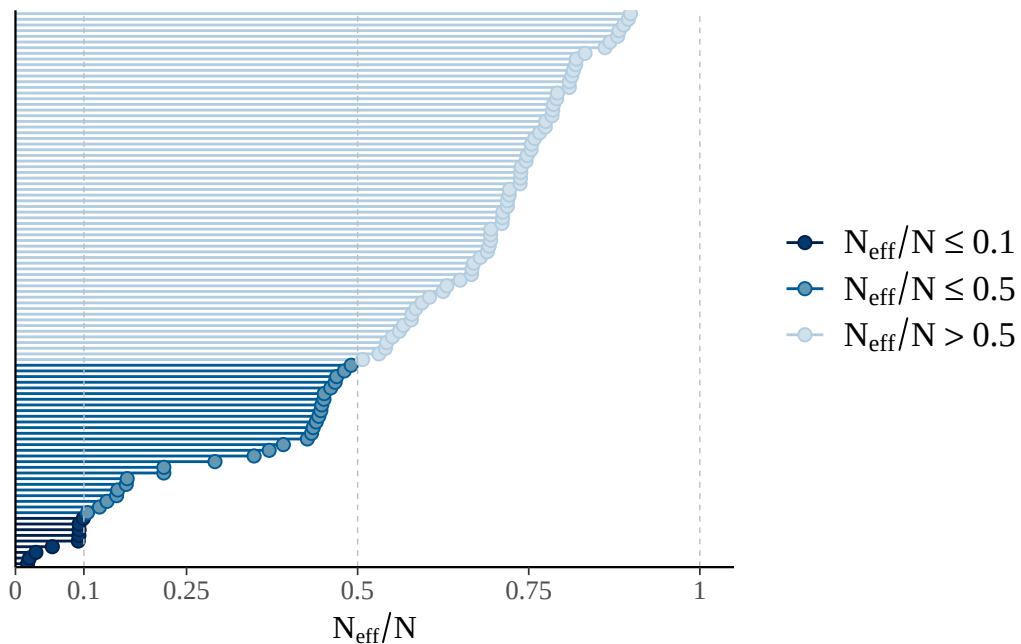
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01769	0.44548	0.67429	0.71194	1.05774	1.92376

[1] "n_eff ratio tail ESS:"

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.09336	0.54031	0.67816	0.61001	0.76043	0.89860

[1] "effective sample size in bulk or tail too small"

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



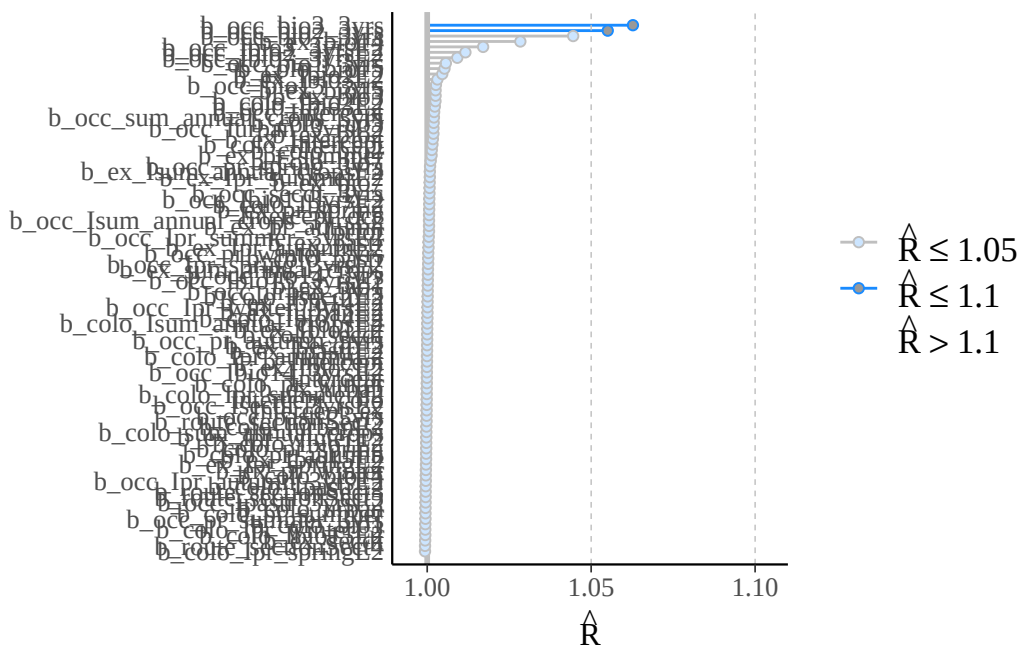
Post-warmup iteration



Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9993	0.9998	1.0002	1.0028	1.0013	1.0627

```
[1] "R-hat > 1.05 for 2 parameters."
```



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_pr_spring_3yrs"
[3] "b_occ_pr_autumn_3yrs" "b_occ_Ibio1_3yrsE2"
[5] "b_occ_Ipr_spring_3yrsE2" "b_occ_Ipr_summer_3yrsE2"
[7] "b_occ_Isecdf_3yrsE2"
```

```
[1] "b_colo_bio1"          "b_colo_pr_winter"
[3] "b_colo_Ibio1E2"       "b_colo_Ibio2E2"
[5] "b_colo_Ipr_springE2"  "b_colo_Ipr_summerE2"
[7] "b_colo_Ipr_winterE2"  "b_colo_sum_annual_crops"
[9] "b_colo_secdf"         "b_colo_pastr"
```

[11]	"b_colo_urban"	"b_colo_Isum_annual_cropsE2"
[13]	"b_colo_IsecdfE2"	"b_colo_IpastrE2"
[1]	"b_ex_Intercept"	"b_ex_bio2"
[3]	"b_ex_bio7"	"b_ex_pr_summer"
[5]	"b_ex_Ibio7E2"	"b_ex_Ipr_springE2"
[7]	"b_ex_Ipr_summerE2"	"b_ex_sum_annual_crops"
[9]	"b_ex_secdf"	"b_ex_pastr"
[11]	"b_ex_Isum_annual_cropsE2"	

